

A balancing-selected chromosome-2 supergene and a neural–photoreceptive program of recent selection in the northern shortfin squid *Illex illecebrosus*

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Abstract

Chromosomal inversions are increasingly recognised as engines of adaptation, yet for most wild inversions we know neither their age, their frequency history, nor whether they are held by selection at all. The northern shortfin squid *Illex illecebrosus* — an annual, semelparous, broadcast-spawning cephalopod with one of the largest effective population sizes of any animal — was recently shown to be panmictic across its northwest Atlantic range (Baker et al., 2025). Using whole-genome data from 633 individuals across that range, we show that this unstructured genome carries a single ~19 Mb inversion on chromosome 2, resolve what the inversion is, and map the genome-wide landscape of recent positive selection around it. The inversion (chr2:60.54–79.50 Mb) segregates at a derived frequency of 0.374 with no geographic cline and between-karyotype $F_{ST} = 0.37$ against geographic $F_{ST} \approx 0$, ruling out local adaptation. It is not neutral: the rare derived arrangement is 2.3-fold more diverse than its frequency can sustain, and a coalescent fit with an ancestral-recombination-graph simulator that enforces the arrangement barrier (msinv) dates it to ~850 ka, reconstructs a long plateau near $p \approx 0.74$, and identifies an active decline to its present frequency over the last ~50–100 ky ($N_e s \sim 10^2$ – 10^3), a history that neutral drift cannot produce. The block captures a gene-poor, functionally unremarkable cargo (no gene-ontology enrichment against a permutation null) yet accumulates 3,513 near-fixed differences whose coding fraction is concentrated in nine genes — the architecture of a balancing-selected supergene. Genome-wide, a stratified, cross-method-corroborated sweep scan built for a strongly expanding, unphased genome recovers 662 Tier-1 and 34 high-confidence Tier-2 sweeps that resolve into two coherent themes, **neurodevelopment** and **photoreception**: screening for background selection sharpens rather than erodes them (response to light 1.9 → 3.1-fold, response to radiation 1.8 → 2.8-fold, neuron-projection and cell morphogenesis 1.7–1.8 → 2.2–2.3-fold). Alongside a cephalopod-lineage-specific gene under a hard sweep and a Z-linked RNA-regulation/hormone-signalling cluster, these results portray a large- N_e , expansion-dominated genome in which a single balancing-selected supergene structures the population while recent directional selection has targeted the sensory and neural program of a fast, migratory, visually guided predator.

1 Introduction

Chromosomal inversions suppress recombination between arrangements and thereby bind together the alleles they capture, allowing co-adapted combinations to persist as segregating “supergenes”

(Kirkpatrick, 2010; Schwander et al., 2014; Thompson and Jiggins, 2014; Wellenreuther and Bernatchez, 2018; Faria et al., 2019). The best-understood cases are those with visible phenotypes and clear selective regimes: mimicry in *Heliconius* (Jay et al., 2021), alternative reproductive morphs in the ruff (Lamichhaney et al., 2016), colony organisation in fire ants (Wang et al., 2013), life-history transitions in *Mimulus* (Lowry and Willis, 2010). The theory that explains why inversions spread is correspondingly well developed for one scenario — local adaptation with migration, in which an inversion is favoured because it protects locally adapted alleles from recombination with immigrant haplotypes (Kirkpatrick and Barton, 2006; Charlesworth and Barton, 2018). Far less is known about inversions that segregate at intermediate frequency *without* geographic structure, where the maintaining force must be some form of balancing selection — overdominance, associative overdominance generated by the recessive deleterious load that accumulates in a non-recombining block (Zhao and Charlesworth, 2016; Gilbert et al., 2020; Jay et al., 2021), negative frequency dependence, or temporally fluctuating selection (Wittmann et al., 2017; Rudman et al., 2022). For such inversions the key evolutionary quantities — age, historical frequency trajectory, and the sign and strength of selection acting on the arrangement — are rarely estimated, largely because the coalescent inside an inversion is not that of a freely recombining locus: arrangements are partially isolated demes exchanging ancestry only by rare double crossovers or gene conversion, and their diversity and divergence must be modelled with the barrier explicitly in place (Navarro et al., 2000; Guerrero et al., 2012; Berdan et al., 2023).

Cephalopods offer an unusual setting in which to ask these questions. They combine an extreme life history — semelparity, enormous fecundity, near-total early mortality, and boom–bust recruitment — with population sizes among the largest of any animal, and with the most elaborate nervous systems and visual systems in the invertebrates (Albertin et al., 2015, 2022; Styfals et al., 2022; Songco-Casey et al., 2022). Their genomes show extensive lineage-specific reorganisation (Schmidbaur et al., 2022) and pervasive RNA-editing-based transcriptome plasticity (Liscovitch-Brauer et al., 2017), and their photoreception extends beyond the camera eye to dermal opsins and light-driven chromatophore control (Mäthger et al., 2010; Ramirez and Oakley, 2015; Kingston et al., 2015). The northern shortfin squid *Illex illecebrosus* is an annual, semelparous broadcast spawner distributed across the northwest Atlantic shelf and slope, performing diel vertical migrations through a steep light gradient and supporting large commercial fisheries (O’Dor and Dawe, 1998; Arkhipkin et al., 2015). Its fecundity-with-high-mortality demography is the profile that motivates sweepstakes-reproductive-success models, under which a few individuals contribute disproportionately to each generation and the genealogy departs from the standard Kingman coalescent (Hedgecock and Pudovkin, 2011; Eldon and Wakeley, 2006; Árnason et al., 2023); a very large effective size, meanwhile, sharpens the resolution of both demographic inference and selection scans.

A recent population-genomic survey of *Illex illecebrosus* (Baker et al., 2025) found no genetic structure across more than 2,500 km of range and no evidence of large structural variants, and recommended management as a single stock. We confirm the panmixia here, but find beneath it a single, striking axis of structure on chromosome 2 — recovered by PCA, F_{ST} , and windowed local PCA and attributable to a large inversion — that raises the questions motivating this study. First, *what is* the chromosome-2 inversion — a neutral, drifting polymorphism or a selected supergene, and if selected, of what kind, what age, and with what frequency history? Second, *where else* in the genome has recent positive selection acted, and on what biology? A differentiation scan cannot answer either: in a population with no geographic structure it has nothing to differentiate, and the sweeps of a large- N_e species leave their footprint in diversity and the site-frequency spectrum rather than in F_{ST} . Underpinning both questions is a third: whether *Illex illecebrosus*’s demography can be distinguished from the multiple-merger coalescent that its reproductive mode predicts, since

every age and selection estimate that follows is licensed by that answer.

We address these with whole-genome data from 633 *Illex illecebrosus* individuals spanning eleven fishery-management divisions, supported by an *I. coindetii* outgroup for divergence-based inference and by *I. argentinus* as a recently diverged congener. We dissect the chromosome-2 inversion with per-arrangement diversity and divergence, a purpose-built barrier-aware coalescent simulator, and a direct characterisation of its gene content and fixed-difference architecture; we infer demography from the site-frequency spectrum and linkage disequilibrium and test the coalescent class explicitly; and we scan for selective sweeps with a framework designed for the failure mode that strongly expanding, unphased, missing-data-rich genomes impose on simulation-trained classifiers (Kern and Schrider, 2018; Johri et al., 2022). The scan resolves into two coherent biological themes — neurodevelopment and photoreception — that strengthen, rather than dissolve, when we control for background selection.

2 Results

2.1 Sample, sex, and the absence of geographic structure

We analysed 633 *Illex illecebrosus* individuals from eleven NAFO management divisions spanning ~ 37 – 50°N (fig. 1, table 1). Read coverage on the Z chromosome separates the sample into 342 ZZ males and 291 ZO females, consistent with a ZZ/ZO system in which females are the heterogametic sex with a single Z and no W (fig. 2); a chromosome previously suspected to be sex-linked (chr42) is autosomal by this criterion. Autosomal analyses use both sexes; the Z, on which males are diploid and females haploid, is analysed separately (section 2.5).

Consistent with the range-wide panmixia reported by Baker et al. (2025), population structure exists in exactly one place in the genome, and that place is not geographic. A windowed principal-component analysis along chromosome 2 localises the three-cluster structure to the inversion (among-cluster variance $\eta^2 = 0.748$ inside it against $\eta^2 = 0.003$ in collinear sequence; Supplementary Fig. S3), and two genome-wide tests show it is unique. First, the chromosome-2 karyotype explains none of the variation on any other chromosome (per-chromosome PCA coloured by karyotype: $\eta^2 = 0.76$ on chr2 versus ≤ 0.01 on all 45 others). Second, and independently of the chr2 labels, a Gaussian-mixture screen of each chromosome’s leading principal component supports more than one cluster only on chromosome 2 (three clusters, $\Delta\text{BIC} = 199$, cluster separation 8.6σ) and on the Z (two clusters — the expected ZZ/ZO hemizygosy split); every autosome is a single cloud (fig. 2). The residual dispersion on minor PCA axes tracks neither division, latitude, nor coverage (Supplementary Fig. S1), and a KING-robust relatedness screen (Manichaikul et al., 2010) finds no duplicate, parent–offspring, or full-sib pairs and a single second-degree pair among the 633 individuals (maximum kinship 0.16): despite sweepstakes reproduction, the sample behaves as effectively unrelated. Because a per-SNP differentiation scan is the most sensitive test for local adaptation in a population this panmictic, we compared per-SNP F_{ST} among divisions to a coalescent neutral null — a single panmictic population simulated under the fitted demography at the empirical sample sizes and missingness. Across the full callsets (551,344 SNPs in 12,007 genes) the empirical differentiation lies entirely within the drift expectation: no SNP exceeds the neutral maximum, a single intergenic site passes the lenient threshold against ≈ 556 expected by chance, and there are zero genic outliers (Supplementary Figs. S2, S19). Nucleotide diversity and Tajima’s D are flat across latitude and pairwise F_{ST} among divisions is ≈ 0 throughout (fig. 6). There is therefore no geographic local adaptation anywhere in the genome, and the inversion is geographically non-clinal — which reframes it from “the thing that structures the population” into an evolutionary object to be characterised in its own right.

2.2 A large, balancing-selected inversion on chromosome 2

Identity, breakpoints, and karyotypes. Local PCA (Li and Ralph, 2019) and a Gaussian-mixture karyotype assignment (95.3% confident) resolve three genotype clusters on chromosome 2 — 254 AA, 284 AB, and 95 BB individuals (fig. 3a) — whose differentiation is karyotypic, not geographic: $F_{ST}(AA, BB) = 0.37$ inside the inversion against a geographic $F_{ST} \approx 0$. Between-karyotype F_{ST} computed along the whole chromosome is ≈ 0 (median 0.006) across collinear sequence and jumps to a median of 0.34 (maximum 0.53) inside the inverted body (Supplementary Fig. S20), so the karyotypes are one panmictic population everywhere except within the barrier. The three karyotypes co-occur throughout the range with no latitudinal cline in arrangement frequency (0.29–0.46 across divisions; fig. 1; Supplementary Fig. S6 and Table S2). Two independent lines pin the breakpoints to **chr2:60.54–79.50 Mb** (a ~ 18.96 Mb differentiated body): a step-change in arrangement differentiation and a strand flip in a whole-chromosome alignment to *I. coindetii* (Supplementary Fig. S11); an extended block of linkage disequilibrium coincides with the same interval (Supplementary Fig. S4).

Polarisation. The derived, inverted arrangement is **BB, at a present frequency of 0.374**; AA is ancestral and standard. This is the reverse of the naive polarisation from the reference allele, which is circular because the *Illex illecebrosus* reference assembly is itself in the inverted orientation (it carries BB at 86.5% of fixed-different sites). Restricting to the non-circular subset of sites at which *I. coindetii* carries the *Illex illecebrosus* alternate allele shows AA to be ancestral (63.0%) and BB derived — a single inversion event, with the reference simply assembled from an inverted individual (Supplementary Table S3).

The inversion is not neutral. The primary evidence is per-arrangement diversity (table 2, fig. 3c). Inside an inversion the two arrangements behave as partially isolated demes (Navarro et al., 2000; Guerrero et al., 2012), so at neutral equilibrium the diversity of each should scale with its frequency. The rare derived arrangement is instead *more* diverse than its frequency can sustain: $\pi_I/\pi_S = 1.356 \pm 0.048$ against an equilibrium expectation of $p/(1-p) = 0.60$ for a polymorphism held at 0.374 — a 2.3-fold excess that is the signature of an old polymorphism that was once common, not of a recent rise. Two mutation-rate-free diffusion arguments exclude neutrality. *Attainment*: reaching the inferred historical peak frequency of ~ 0.74 by drift takes at minimum ~ 1.15 million generations at the binding (smallest) N_e — $1.36\times$ the entire inferred lifetime of the inversion, and up to $17\times$ at present N_e . *Decline*: the observed fall from 0.74 to 0.374 is 6.7 standard deviations of neutral drift at the fitted duration, and a coalescent profile of the fall duration excludes the slow, drift-compatible alternative at $\Delta\chi^2 \gtrsim 8-80$ (Supplementary Fig. S14). The arrangement was actively removed, not lost to drift.

Age and frequency history from a barrier-aware coalescent. Estimating an inversion’s age requires a coalescent in which the two arrangements exchange ancestry only across the barrier; standard simulators do not provide this. We therefore fit the inversion’s age and frequency trajectory with **msinv** (Small, 2025), an ancestral-recombination-graph simulator that enforces the cross-karyotype recombination barrier, tracks ancestral material with a hull-based scheme in the msprime lineage (Kelleher et al., 2016; Baumdicker et al., 2022), and outputs tree sequences (section 5). The data constrain the *shape* of the frequency trajectory $p(t)$ rather than any site-level mechanism, so we condition on a deterministic trajectory under within-arrangement neutrality and fit jointly to π_I/π_S , d_{xy}/π_I , and the per-arrangement singleton spectrum (fig. 4b; Supplementary Figs. S5, S14). The coalescent age is ~ 850 ky. It is the robustly determined quantity because it

is carried by d_{xy} between arrangements, which is symmetric to the polarisation — the age survived the complete reversal of the arrangement polarity described above, a check few inversion ages can claim. The reconstructed history is that the inversion arose ~ 850 ka, coincident with the onset of the population expansion (section 2.3); was maintained near $p \approx 0.74$ for most of its history; and then fell sharply to 0.374 over ~ 50 – 100 ky, ending $\gtrsim 50$ – 100 ka, under selection of order $N_e s \sim 10^2$ – 10^3 ($s \approx 1.5 \times 10^{-4}$). Simulated inversions of 37 kb rescaled to the 19-Mb body reproduce these statistics with length-invariance verified over a 25-fold range, and the fitted model reproduces the observed joint two-dimensional site-frequency spectrum between arrangements — near-fixed differences and off-diagonal density in the inverted body, collapse to the diagonal in collinear controls (Supplementary Fig. S5). Every absolute age scales inversely with the mutation rate $\mu = 3 \times 10^{-9}$, the dominant external uncertainty.

What the inversion is not. It is not a region of locally suppressed crossing-over: recombination inside the body is at the chromosomal background rate by direct measurement (the ~ 19 kb LD windows a recombination-map method reads are blind to a 19-Mb barrier; Supplementary Fig. S12), so the differentiation is a consequence of the arrangement barrier itself. It is not locally adaptive: there is no cline and geographic $F_{ST} \approx 0$, so the migration–selection scenario that explains most well-studied inversions (Kirkpatrick and Barton, 2006) cannot apply. And it is not a swept region: the genome-wide scan below finds *zero* cross-method-concordant sweeps on chromosome 2 — the recombination-map method is broadly elevated and the classifier over-calls across the inversion, but the two never co-localise into focal concordance, exactly the pattern expected for a balancing-selected supergene rather than a swept locus, and correctly filtered by the concordance requirement. Gene flux between arrangements is not detectable in mean d_{xy} , though a residual $\sim 40\%$ depletion of near-fixed differences in the outermost megabase remains after the breakpoints are pinned. The mechanism of balancing selection — overdominance, associative overdominance, frequency dependence, or a moving optimum under fluctuating selection — is not resolved by these data; a shelf-spawning squid whose paralarvae ride the Gulf Stream is a plausible setting for temporally fluctuating selection on ~ 100 ky glacial cycles, but this remains a hypothesis.

Gene content and the architecture of arrangement divergence. The block has no functionally coherent cargo, yet it carries the molecular signature of a young, recombination-suppressed supergene. It is gene-poor — 89 gene models, 45 with cross-species functional annotation, against a median of ~ 50 annotated genes in random windows of the same size — and functionally scattered: lipid metabolism, DNA repair, transcription and splicing, signalling, with no coherent pathway (Supplementary Table S7). A gene-ontology test of its annotated genes against the genome returns *no* enriched term at $FDR < 0.05$ in any ontology, and a permutation null shows this is genuine rather than a power artefact: random gene sets of the same size produce *more* nominal hits than the inversion genes (86.6 ± 60 vs 69; permutation $p = 0.51$), and the eye-catching ~ 40 -fold per-term enrichments rest on as few as two genes (Supplementary Fig. S21). A previously reported metabolic enrichment does not survive this null. The inversion has thus captured a functionally unremarkable slice of the genome, not a selected gene set — consistent with a supergene maintained for the arrangement as a whole. What the arrangements *do* accumulate is divergence-as-a-unit: 3,513 near-fixed differences between AA and BB (per-arrangement $|\Delta AF| \geq 0.9$) are spread across the body (Supplementary Fig. S5). These are overwhelmingly non-coding — coding sequence is only $\sim 0.2\%$ of the region — but the exonic fixed differences are ~ 2.3 -fold over-represented relative to that footprint and are strikingly *clustered*: all 19 fall in just nine genes, led by a cytochrome P450 (*CYP4X1*) and an acidic-repeat protein (four each) and including a phospholipase A2, the ciliary

gene *RPGRIP1L*, and a cysteine-rich venom-like protein. Although no locus in the block shows a classical sweep and the region is enriched for no function, the coding divergence that recombination suppression shelters is concentrated in a handful of genes rather than spread evenly — the expected architecture of a balancing-selected inversion that recombines with its homologue only rarely, and the raw material from which a supergene’s mutation load and eventual degeneration are built (Jay et al., 2021; Berdan et al., 2023).

2.3 Demographic history licenses the inversion model

The age and selection estimates above depend on the population’s demography and on the coalescent class being Kingman. Genome-wide, *Illex illecebrosus* shows $\pi \approx 0.0093$ and Tajima’s $D \approx -2$ in essentially every window (Tajima, 1989a,b) — an unambiguous, genome-wide signature of expansion (table 3, fig. 4a, fig. 6). A moments fit to the folded SFS gives growth from an ancestral $N_e = 547,928$ to a present $N_e = 6,808,096$ over 769,519 generations (~ 12 -fold; generations $\approx$ years for this annual species), and this history reproduces both the observed π and a full-SFS Tajima’s $D = -1.92$ under forward simulation. Stairway Plot 2 and GONE2 concur on strong recent expansion to N_e in the $1.5\text{--}1.7 \times 10^7$ range. Because sweepstakes reproduction would mimic expansion by inflating rare variants (Eldon and Wakeley, 2006; Hedgecock and Pudovkin, 2011; Eldon et al., 2015), we tested the coalescent class directly: the folded SFS is fit far better by a Kingman-plus-growth model than by any multiple-merger (Beta) coalescent (Schweinsberg, 2003) (shape deviation $L_1 = 0.036$ vs a best- α 0.108), and contemporary N_e estimates from the SFS and from linkage disequilibrium agree — both inconsistent with a strong multiple-merger process. The expansion began ~ 770 ka, i.e. at essentially the same time as the inversion arose, which places the supergene’s origin at the start of the demographic episode that shapes the whole genome.

2.4 Genomic context: recombination, background selection, and protein adaptation

Three further features of the genome set the stage for the sweep scan. *Recombination*, as estimated by a deep-learning LD-based map built per sex, is $\sim 0.21\text{--}0.24$ cM/Mb on the autosomes (sex-averaged ~ 0.244 ; Supplementary Fig. S12); because the estimator reads linkage disequilibrium it recovers the sex-averaged rate and heterochiasmy is not identifiable from it (the female-only map in fact collapses to a near-constant and is uninformative). The map is flat with respect to gene content — the rate varies by only $\sim 6\%$ between windows adjacent to exons and windows >100 kb from any exon, unchanged when π is held fixed (Supplementary Fig. S17a) — as a crossover rate should be. It is also compressed at chromosome scale: on chromosome 1 the rolling median spans just $0.17\text{--}0.23$ cM/Mb, the pericentromeric flanks where π halves show no more than a 5% dip, and the centromere itself (a 2.5-Mb block that is 80% repeat and almost entirely masked) carries no windows (Supplementary Fig. S22). The pattern is genome-wide: across all 43 scanned autosomes the chromosome-scale rolling median varies by only 10–41% of each chromosome’s median, and no chromosome shows a peak more than 25% above it. The map therefore fixes the genome-wide mean well but resolves little broad-scale landscape, and “nearly uniform” is in part an estimator ceiling; none of the analyses below depend on fine-scale recombination variation. *Background selection* is moderate and gene-density-driven: π falls with local coding density ($r = -0.31$; 20–30% reduction in gene-dense windows) and rises 43% between exon-adjacent and exon-distant windows, while recombination does neither (Supplementary Fig. S17) — the classical footprint of linked purifying selection (Charlesworth et al., 1993; McVicker et al., 2009; Charlesworth and Jensen, 2021), organised here by gene density rather than by the recombination axis. Diversity is also heterogeneous at

the megabase scale: 22 blocks of ≥ 1 Mb on 17 chromosomes carry $2\text{--}4\times$ the chromosome-median π with less-negative Tajima’s D (e.g. chr1:24–31 Mb, $\pi = 0.020$; Supplementary Fig. S22c). We tested these against every structural explanation and they fail all of them: local PCA finds no karyotype clusters (unlike the chr2 positive control), there is no heterozygosity excess, no excess-heterozygosity filtering loss, and no per-individual depth anomaly, and their allele-frequency spectrum is identical to flanking sequence — they simply carry $\sim 2.5\times$ more SNPs. They are gene-poor (coding fraction 0.001), largely unalignable to *I. coindettii* (35% vs $\sim 88\%$ of bases aligned) and 1.7-fold more divergent where alignable, and low-coverage *I. argentinus* reads also call $\sim 3\times$ more variants in them. They are therefore fast-evolving, largely lineage-specific gene deserts — elevated mutation input and relaxed linked selection — not structural polymorphisms, and the chromosome-2 inversion remains the genome’s only segregating karyotype. A model-based B -map could not be fit under this regime (the classical approximation overflows where the deleterious mutation rate exceeds the selection coefficient), so local coding fraction serves as the background-selection covariate that the sweep scan controls for explicitly. *Protein adaptation* over the divergence timescale is substantial: McDonald–Kreitman tests (McDonald and Kreitman, 1991; Messer and Petrov, 2013a) against *I. coindettii* across 20,999 genes give a genome-wide proportion of adaptive substitutions $\alpha \approx 0.34\text{--}0.64$, with the splicing factor *SF3B4* the single most significantly adaptive gene (Supplementary Fig. S10). Per-gene power is limited, so we treat the MK results as complementary to the recent-sweep scan, which detects a different — recent, often non-coding — timescale of selection. *I. argentinus* is a striking comparator: recently diverged (an order of magnitude fewer fixed coding differences than *I. coindettii*; Supplementary Table S3), it carries near-identical genome-wide diversity ($\pi = 0.0092$ vs 0.0093, within 1%) — two large- N_e , expansion-shaped congeners of matched standing variation.

2.5 A genome-wide scan for selective sweeps

We scanned for recent positive selection with diploS/HIC (Schrider and Kern, 2016; Kern and Schrider, 2018), a five-class site-frequency/diversity-based classifier (neutral, hard, soft, linked-hard, linked-soft; 1.1 Mb feature windows of eleven 100 kb sub-windows) trained on forward and coalescent simulations under the fitted growth demography (fig. 5; Supplementary Figs. S7, S8).

The scan must be framed as an outlier detector. Two methodological findings shape everything that follows. First, a training/scanning feature mismatch — the model was initially trained on phased-haplotype simulation features but applied to unphased genotypes — inflated linkage-based features by an order of magnitude and had to be corrected by retraining on the same unphased feature extractor used to scan; the final full-SFS model reaches 81% class accuracy (Supplementary Fig. S7). Second, and more fundamentally, even the corrected model *over-calls* sweeps on the real genome: real neutral windows (accessible, ≥ 5 kb from any gene) carry 3–8-fold more within-window spatial heterogeneity in π and θ than the simulations, even though their marginal SFS matches ($D \approx -2$). We ruled out missing-data structure and sample incompleteness as the cause; the residual is real per-sub-window diversity variation (mutation-rate variation plus background selection) that a uniform- θ coalescent does not reproduce — a concrete instance of the simulation-to-reality gap that current best-practice guidance warns against (Johri et al., 2022). We therefore use the classifier as a **stratified empirical-outlier detector**: the sweep score ($p_{\text{hard}} + p_{\text{soft}}$) is ranked within strata of callable fraction and coding density, so a window is an outlier only relative to windows of similar accessibility and gene density, and calls are corroborated by an independent LD-footprint method (RAiSD; Alachiotis and Pavlidis 2018) or a composite-likelihood SFS method (SweepFinder2; Nielsen et al. 2005; DeGiorgio et al. 2016). We also show that an inherited minor-allele-frequency filter was a mistake under strong expansion (it discards the rare-variant excess that

carries both the expansion and the sweep signal) and use the full spectrum on both the simulation and empirical sides.

Two tiers of sweep calls. We report sweeps at two levels of stringency (Supplementary Fig. S13). **Tier 1** is the diploS/HIC hard/soft classification, spatially resolved by a recombination-aware Markov decoder — a hidden Markov model over the per-window (100 kb) classes, with neutral, linked, and sweep-centre states whose emissions are the prior-corrected class posteriors and whose dwell lengths follow the ~ 400 kb sweep footprint scaled by the recombination map (Supplementary Fig. S13a). Because the genome reads as $\sim 91\%$ linked-selected under strong expansion, the decoder’s role is spatial coherence: it retains hard/soft calls that lie in contiguous sweep centres and discards spatially isolated ones, reducing 1,758 raw calls to 717 windows (658 localised autosomal calls; 459 hard, 199 soft) and preferentially keeping the higher-outlier windows (median empirical-outlier percentile 0.98 vs 0.95 for discarded calls; Supplementary Fig. S13b). **Tier 2** is the subset independently corroborated by a footprint method: the 34 cross-method-concordant regions (a diploS/HIC-HMM outlier plus RAiSD or SweepFinder2). The two tiers agree strongly — 89% of windows at concordant loci are decoded as sweep centres versus 2.9% genome-wide (odds ratio 282, $p < 10^{-46}$), and 30 of the 34 concordant regions contain a Tier-1 centre (Supplementary Fig. S13c) — so the conservative concordance set is a high-purity subset of the classifier calls, not a disjoint list. Because the decoder could not be simulation-calibrated at this genome’s recombination density (section 5), its dwell parameters are set from the sweep footprint and the recombination map and validated post hoc by this concordance; we treat Tier 1 as a ranked candidate set and Tier 2 as the high-confidence calls. The Z chromosome, which the footprint methods do not cover, is necessarily diploS/HIC-only; scanned separately in males (below), it contributes four further Tier-1 calls, for a genome-wide Tier-1 total of 662 (fig. 5; Supplementary Fig. S18).

Two themes: neurodevelopment and photoreception. The 658 autosomal Tier-1 calls overlap 894 genes (476 with GO annotation; one-third of calls fall in gene-poor, often transposon-rich windows and are excluded). A gene-ontology enrichment against the annotated-gene background (Fisher’s exact test, Benjamini–Hochberg FDR; Benjamini and Hochberg 1995) returns 124 terms at $\text{FDR} < 0.05$ (Supplementary Fig. S15, Table S4), dominated by **neural development** — neurogenesis, generation of neurons, neuron development, neuron-projection and cell morphogenesis (1.5–1.8-fold) — together with transcriptional and RNA-regulatory control, and a weaker **light-response** signal (response to light stimulus 1.9-fold, response to radiation 1.8-fold). The enrichment is carried by the hard-sweep calls (99 terms); the disjoint soft-sweep set has none at $\text{FDR} < 0.05$. The decisive test is whether these themes are artefacts of background selection — gene-dense neural loci have lower diversity for reasons unrelated to sweeps. We therefore screened every Tier-1 call with the same per-region background-selection test used for the concordant set, retaining only calls whose local π falls below the background-selection expectation given flanking coding density. This keeps 291 of the 658 (237 hard, 54 soft), and the enrichment does not merely survive the screen — it *sharpens*. The BGS-robust set yields 152 terms at $\text{FDR} < 0.05$, and every leading term’s fold enrichment rises: response to light stimulus $1.92 \rightarrow 3.12$ ($\text{FDR } 2 \times 10^{-2} \rightarrow 1 \times 10^{-4}$), response to radiation $1.75 \rightarrow 2.77$ ($\text{FDR } 3 \times 10^{-2} \rightarrow 2 \times 10^{-4}$), neuron-projection morphogenesis $1.81 \rightarrow 2.26$, cell morphogenesis $1.72 \rightarrow 2.24$, neuron development $1.59 \rightarrow 1.97$, generation of neurons $1.60 \rightarrow 1.90$, transcription by RNA polymerase II $1.73 \rightarrow 2.08$ (Supplementary Fig. S15). The photoreception signal — weakest in the full set — becomes the single most enriched theme after the screen, and its more specific child terms (detection of light stimulus $2.5 \rightarrow 3.9$, phototransduction $2.5 \rightarrow 3.4$) move in the same direction, though with too few genes to reach significance.

A shared-gene enrichment map of the BGS-robust set resolves the same two connected modules, neural development/morphogenesis and light response, with no third (Supplementary Fig. S16, Table S5). Because background selection can only remove diversity where genes are dense, a signal that grows as gene-dense false positives are removed is the opposite of what a BGS artefact would produce: the sweep program in *Illex illecebrosus* is a coherent sensory-and-neural one, and it is real.

Candidate sweeps. The Tier-2 concordant regions carry a biologically coherent set of cephalopod-orthologous genes that sit squarely within these themes (table 4; Supplementary Table S6): the neural/EMT transcription factor *ZEB2*, the synaptic-vesicle GTPase *Rab3*, the apelin receptor *Aplnr* (cardiovascular/vasculature signalling), the DNA-damage-response gene *MACROD2*, the phospholipid synthase *cept1*, and — at a locus the annotation had missed — a ceramide phosphoethanolamine synthase. A per-region background-selection test isolates 11 of these as high-confidence candidates in which diversity is reduced beyond the background-selection expectation, and fine-scale localisation confirms all 11 as recent sweeps by the sign of Tajima’s *D* at the diversity valley, reassigning two apparent gene-desert signals to transposable-element regions (flagged lower confidence) and one candidate’s target to a neighbouring gene. The single most distinctive autosomal candidate is a **cephalopod-lineage-specific gene under a hard sweep** on chromosome 16 — a tandem-exon transcript with no functional annotation whose translation matches an uncharacterised protein family in the bobtail squid *Euprymna* ($E = 2 \times 10^{-53}$) — the kind of lineage-specific novelty that cephalopod genomes are known to generate (Albertin et al., 2015, 2022). There is no overlap between the sweep candidates and the MK-adaptive genes, as expected for their different timescales.

The Z chromosome. Because males are Z-diploid and females Z-haploid, the Z was scanned separately in the 330 confident ZZ males, using a model retrained at the matched sample size (the classifier’s features are sample-size dependent) and a formal adequacy gate for the Z’s reduced effective size; the male-only screen was validated against the 289 haploid females, whose per-window diversity tracks the males’ at $r = 0.996$. Of the four Z Tier-1 calls (three hard, one soft; Supplementary Fig. S18), three credible, gene-localised sweeps survive fine-scale triage, all confirmed as conserved cephalopod genes by direct homology: **Pasilla/Nova-1** (a neuronal alternative-splicing regulator; 92% identity to *Octopus*), **Smaug/samd4a** (a germline RNA-binding translational repressor), and **Esrrg** (an estrogen-related receptor with an adjacent likely-*cis*-regulatory lncRNA; 97.5% identity to *Octopus*) — a coherent RNA-regulation-and-hormone-signalling signature on the sex chromosome that echoes the neural theme of the autosomes. The *Amnionless*/"Zfest" sex-determination locus falls just outside the swept valley and shows only purifying selection: a linked passenger, not the target.

3 Discussion

Illex illecebrosus emerges as a large- N_e , expansion-dominated genome carrying a single dramatic axis of structure — a balancing-selected chromosome-2 supergene — against an otherwise unstructured, moderately background-selected background on which recent directional selection has acted on a specific biology. Where the earlier survey (Baker et al., 2025) established panmixia for management, our contribution is to find the inversion beneath it, to explain it, and to look past it.

The inversion is a datable, declining, balancing-selected supergene. Most inversions whose evolutionary dynamics are understood are maintained by local adaptation with migration (Kirkpatrick and Barton, 2006; Lowry and Willis, 2010; Faria et al., 2019); the *Illex illecebrosus* inversion cannot be, because there is no geography to adapt to — arrangement frequency is non-clinal and geographic F_{ST} is zero everywhere. It must instead be held by balancing selection, and the data constrain *how* it has been held with unusual precision. The excess diversity of the rare derived arrangement forces a history in which BB was once common; two mutation-rate-free arguments and a barrier-aware coalescent fit converge on an ~ 850 ky-old polymorphism that plateaued near 0.74 and has been actively driven down to 0.37 over the last ~ 50 –100 ky. This is a qualitatively different object from the classic *Drosophila* inversions, whose frequencies track clines and seasons (Corbett-Detig and Hartl, 2012; Kapun and Flatt, 2019), and from the human 17q21 inversion, whose signature is a recent rise under positive selection (Stefansson et al., 2005). It resembles more the mimicry supergene of *Heliconius numata*, where an inverted arrangement carries a recessive deleterious load that generates associative overdominance and caps its frequency (Jay et al., 2021). Associative overdominance is a natural candidate here: theory shows it emerges precisely in large, non-recombining blocks once recessive load accumulates (Zhao and Charlesworth, 2016; Gilbert et al., 2020; Berdan et al., 2021), and the clustered coding divergence we document — 19 exonic fixed differences in nine genes, in a block whose 45 annotated genes show no functional enrichment — is the raw material of such a load. But a two-phase history of plateau followed by decline is also what selection toward a shifting optimum can produce (Matuszewski et al., 2015); fluctuating selection is demonstrably strong in wild populations (Wittmann et al., 2017; Rudman et al., 2022), and a shelf spawner whose paralarvae are advected by the Gulf Stream is a plausible target of ~ 100 ky climate cycles. The data constrain the trajectory, not the mechanism. What they do establish is that the arrangement itself, not any single gene within it, is the unit of selection: the block is gene-poor, functionally unremarkable, differentiated as a unit, and free of sweeps, exactly as expected when recombination suppression is the trait under selection (Kirkpatrick, 2010; Berdan et al., 2023).

Two features of the inversion analysis are methodologically general. First, the age is carried by the polarisation-symmetric divergence between arrangements, so it survived a complete reversal of which arrangement is derived — a reversal forced on us by discovering that the reference genome is itself inverted. Reference-orientation circularity is a hazard for any inversion study that polarises from the reference allele, and outgroup-based polarisation on non-reference sites is the remedy. Second, the coalescent inside an inversion is a two-deme process with a barrier (Navarro et al., 2000; Guerrero et al., 2012), and fitting it requires a simulator that enforces the barrier in the ancestral recombination graph rather than approximating the arrangements as freely recombining. msinv provides this, and the plateau-and-decline history it recovers is not accessible to methods that treat the inversion as a single locus. As the catalogue of wild inversions grows (Wellenreuther and Bernatchez, 2018; Mérot et al., 2020), joint estimation of age, trajectory and selection from per-arrangement diversity, divergence and the joint SFS offers a route to classifying them by history rather than by phenotype.

The sweep program is neural and photoreceptive, and it is robust to background selection. The genome-wide scan resolves into two themes. The first — neurogenesis, neuron development, and neuron-projection and cell morphogenesis — is what one might expect of a lineage whose defining evolutionary innovation is a large, centralised, cell-type-rich nervous system (Albertin et al., 2015; Styfahls et al., 2022), and whose genomes are marked by regulatory reorganisation around neural genes (Schmidbaur et al., 2022). The second, photoreception, is the more surprising and

the more *Illex illecebrosus*-specific. Cephalopod photoreception is not confined to the camera eye: opsins and phototransduction cascades are expressed in skin and chromatophores (Ramirez and Oakley, 2015; Kingston et al., 2015), and the visual system is highly cell-type-diverse, with optic lobes elaborated convergently with those of vertebrates (Songco-Casey et al., 2022; Styfals et al., 2022; Yoshida et al., 2015). *Illex illecebrosus* is a fast, visually guided predator that migrates vertically each day through a steep light gradient and seasonally across latitude (O’Dor and Dawe, 1998); selection on light detection is biologically apposite. The decisive evidence that these themes are real, rather than by-products of reduced diversity at gene-dense neural loci, is their behaviour under the background-selection screen. Background selection lowers diversity where genes are dense (Charlesworth et al., 1993; McVicker et al., 2009; Charlesworth and Jensen, 2021); a sweep call that is really a BGS artefact is therefore *removed* by a test that demands diversity below the BGS expectation, and any enrichment riding on such calls should erode. Instead every leading term sharpened — light response $1.9 \rightarrow 3.1$, radiation $1.8 \rightarrow 2.8$, morphogenesis $1.7\text{--}1.8 \rightarrow 2.2\text{--}2.3$ — and the weakest theme became the strongest. The sweep program is not an artefact of gene density.

We frame these as candidates rather than confirmed targets, and the framework itself is offered as a contribution. Simulation-trained per-window classifiers are increasingly standard (Schrider and Kern, 2016; Kern and Schrider, 2018), but on a strongly expanding, unphased, missing-data-rich genome they over-call because real neutral sequence is spatially more heterogeneous than tractable neutral simulations. Rather than hide this behind a threshold, we quantified it and let it dictate the framing — a stratified empirical-outlier scan, cross-method concordance, and an explicit background-selection filter — in the spirit of recent calls for inference that models the confounders it cannot remove (Johri et al., 2022). The predominance of hard over soft sweeps in the enriched set is notable because a population this large is expected to adapt mainly by soft sweeps from standing variation (Messer and Petrov, 2013b); a hard-sweep excess instead points to strong selection on new or rare variants with a small mutational target (Hermisson and Pennings, 2005; Harris et al., 2018), in contrast to humans, where soft sweeps appear to dominate (Schrider and Kern, 2017); the soft-sweep class is underpowered here, however, and we draw no firm conclusion about the mode of adaptation.

Demography, sweepstakes, and what licenses the rest. A sweepstakes-reproduction species is exactly where multiple-merger coalescents are expected to masquerade as expansion (Eldon and Wakeley, 2006; Hedgecock and Pudovkin, 2011), and recent work on Atlantic cod argues that pervasive recurrent sweeps can themselves generate multiple-merger genealogies (Árnason et al., 2023). We tested the coalescent class directly rather than assuming it: the SFS is Kingman-plus-growth, not Beta, and SFS- and LD-based N_e agree. That result is what licenses every age and selection estimate above, and it means that *Illex illecebrosus*’s enormous fecundity has not, at the genomic scale we can resolve, translated into the skewed genealogies its life history would suggest. That the inversion arose at essentially the same time as the expansion began is a coincidence worth noting: a supergene established at the onset of a ~ 12 -fold demographic expansion would have been carried through that expansion at its plateau frequency, and its subsequent decline is a recent event against that backdrop.

4 Limitations

The callset is unphased, so no haplotype-homozygosity statistics are available; sweep detection relies on the SFS and diversity alone, which limits power against soft sweeps in particular. The simulation-to-real neutral mismatch that drives classifier over-calling is real and unresolved at the

level of absolute neutral calling — our empirical-outlier framing is a principled response, not a cure — and it also defeats a within-arrangement SFS approach to the inversion. Every inversion age scales inversely with the mutation rate, and $\mu = 3 \times 10^{-9}$ is the weakest external input; an independent calibration (e.g. a karyotyped congener such as *I. argentinus*) is the clearest route to a mutation-rate-free age. The inversion model conditions on a deterministic frequency trajectory and within-arrangement neutrality; it constrains the shape of $p(t)$, not the mechanism of balancing selection. The LD-based recombination map has a compressed dynamic range at chromosome scale and does not resolve centromeric suppression (Supplementary Fig. S22); it is used only for its genome-wide mean and for the decoder’s dwell scaling, and the absence of a diversity–recombination relationship in these data should be read with that ceiling in mind. The corroborating footprint methods are broad, and the recombination-aware decoder is design-informed rather than simulation-calibrated. Candidate sweep genes rest on homology to distant cephalopod models and on cross-method concordance; the two enrichment themes are statements about gene sets, not about individual confirmed targets.

5 Methods

Full parameter values, software versions, and command-level provenance are in Supplementary Table S1 and in the project repository.

Data. The ingroup is all 633 *Illex illecebrosus* individuals whole-genome sequenced by Baker et al. (2025) (ENA project PRJEB101095; collected 2022–2023 across eleven NAFO divisions, every individual with coordinates), of which that study retained 540 after its own quality filters. We reprocessed the raw reads from scratch: mapping to the *Illex_F24* primary assembly (45 numbered chromosomes plus the Z) and joint variant calling (Supplementary Table S1). Genotypes are unphased and the VCF is variants-only (no invariant sites), so diversity is computed from genotype likelihoods off the BAMs rather than from called genotypes. *I. coindetii* is the divergence/MK outgroup; *I. argentinus* is a secondary comparator. An accessibility mask restricts all analyses to callable sites.

Diversity, SFS, and demography. Site-frequency spectra, Watterson’s θ , per-window π and Tajima’s D , and per-karyotype spectra were computed with ANGSD (Korneliussen et al., 2014) (`-doSaf/realSFS`); per-population statistics were computed per NAFO division and per chromosome (all 45 autosomes plus the Z, the latter in males only). Demographic history was inferred with Stairway Plot 2 (Liu and Fu, 2020), moments (Jouganous et al., 2017), and GONE2 (Santiago et al., 2025), and validated by forward simulation reproducing the observed π and D . The coalescent class was tested by comparing the folded SFS against Kingman-plus-growth and Beta-coalescent expectations. Windowed π , d_{xy} , F_{ST} , and PCA used `pg_gpu` (kr-colab, 2025); per-SNP and between-karyotype F_{ST} used the Hudson estimator (Bhatia et al., 2013). Relatedness was screened with KING-robust (Manichaikul et al., 2010).

Karyotyping and the inversion. Karyotypes were called by local PCA (Li and Ralph, 2019) and a Gaussian-mixture model. Arrangement polarisation used an AnchorWave (Song et al., 2022) *I. coindetii*↔*Illex illecebrosus* whole-genome alignment restricted to non-reference-biased sites, with `est-sfs` (Keightley and Jackson, 2018) as a check. The inversion’s age and frequency trajectory were fit with `msinv` v0.3.5 (Small, 2025), an ARG-based coalescent simulator that enforces cross-karyotype recombination barriers and outputs tree sequences, implementing hull-based ancestral-

material tracking in the msprime lineage (Kelleher et al., 2016; Baumdicker et al., 2022); it was validated against SLiM (Haller et al., 2026), msprime (Baumdicker et al., 2022), and discoal (Kern and Schrider, 2016). All coalescent simulations underlying the inversion age and decline inference were generated with msinv, and the complete chr2-inversion analysis — fitting code, simulation configuration, and every per-analysis output and figure — is housed as a dedicated *Illex illecebrosus* module within the msinv repository (files/illex/, results/illex/). Simulations used the fitted growth demography, $\mu = 3 \times 10^{-9}$, and $r = 1.977 \times 10^{-9}$; statistics were computed on the differentiated body and rescaled from a 37-kb simulated inversion with length-invariance verified over a 25-fold range. Breakpoints were pinned by combining arrangement differentiation with the AnchorWave alignment; missed genes were recovered by DIAMOND blastx (Buchfink et al., 2021) against seven cephalopod proteomes, with transposable elements annotated by Earl Grey (Baril et al., 2024) and transcripts by StringTie2 (Kovaka et al., 2019). Fixed differences between arrangements were defined as sites with $|\Delta A| \geq 0.9$ between AA ($n = 254$) and BB ($n = 95$) homozygotes; the gene-ontology test of inversion genes used Fisher’s exact test against the genome-wide annotated-gene background in all three ontologies, with a permutation null of 500 random gene sets of equal size.

Recombination map. Per-sex recombination maps were built with ReLERNN (Adrión et al., 2020) (SIMULATE→TRAIN→PREDICT→BSCORRECT), dropping individuals with no calls on a chromosome and never forcing diploid genotype calls.

Selective-sweep scan. The classifier was diploS/HIC (Kern and Schrider, 2018) (five classes, 1.1 Mb windows of eleven 100 kb sub-windows, twelve diploid statistics per sub-window), trained on SLiM 5.2 forward simulations (Haller et al., 2026) (recapitated with msprime; Kelleher et al. 2018; Haller et al. 2019) and pure-msprime neutral simulations under the fitted growth demography, at the empirical sample size and with realistic structured missingness. The model was trained on the same unphased feature extractor (fvecVcf) used to scan, on the full spectrum (no MAF filter). It was used as a stratified empirical-outlier detector (score $p_{\text{hard}} + p_{\text{soft}}$ ranked within callable-fraction $\times$ coding-density strata), corroborated by RAiSD (Alachiotis and Pavlidis, 2018) and SweepFinder2 (DeGiorgio et al., 2016). The per-window classifications were spatially resolved into Tier-1 calls with a recombination-aware hidden Markov model (neutral / linked / sweep-centre states; emissions are the prior-corrected collapsed class posteriors; dwell lengths set from the ~ 400 kb sweep footprint scaled by the ReLERNN sex-averaged map, with a constant-rate fallback on chromosomes 2 and 42 and the Z). A simulation calibration of the decoder was attempted but proved infeasible: at N_e -scale recombination the recapitated ancestral recombination graphs carry $\sim 3.4 \times 10^5$ marginal trees per 2.5 Mb, making tree-sequence genotype ascertainment for a training scan prohibitively slow (> 2.7 h per segment); the dwell parameters are therefore design-informed and validated post hoc by the 89%/2.9% Tier-1–Tier-2 concordance (Supplementary Fig. S13). The Z chromosome was scanned in males only with a model retrained at $n = 330$ and passed a comparative feature-adequacy gate, then validated against the haploid females. Candidate genes were annotated with EnTAP and screened by DIAMOND against cephalopod proteomes. Gene-ontology enrichment of Tier-1 genes used Fisher’s exact test against the annotated-gene background on the scanned chromosomes with Benjamini–Hochberg FDR (Benjamini and Hochberg, 1995); the background-selection screen retained calls whose local π fell below the expectation from flanking coding density ($\hat{\pi} = 0.01059 - 0.16547 \cdot f_{\text{CDS}}$; standardised residual ≤ -0.3).

Table 1: **Sample and whole-genome summary.** Diversity statistics are genome-wide from genotype likelihoods; karyotype counts and the inverted-arrangement frequency are for chromosome 2.

Quantity	Value
Species / callset	<i>Illex illecebrosus</i> ; Baker et al. (2025) whole-genome
Individuals (karyotyped)	633
Sex (chrZ coverage)	342 ZZ males / 291 ZO females
Sampling	11 NAFO divisions, $\sim 37\text{--}50^\circ\text{N}$
Geographic structure	$F_{ST} \approx 0$ (none outside chr2)
Genome-wide π	0.0093
Genome-wide Tajima’s D	-2.07 (-2.09 in 50 kb windows)
chr2 karyotypes (AA/AB/BB)	254 / 284 / 95
Inverted (BB) frequency	0.374 (0.29–0.46 across divisions, non-clinal)
Karyotype $F_{ST}(\text{AA, BB})$	0.365

McDonald–Kreitman. MK tests used mkado (Rivera-Colón et al., 2026) against the *I. coindetii* outgroup over 20,999 genes, with degeneracy annotated by degenotate (gwct, 2024).

Data and code availability

The manuscript, all figures and tables, their generating scripts, and editable table exports are in the manuscript repository (https://github.com/stsmall/Illex_popgen); a section-by-section index of every figure and table source file is provided there as FIGURES_AND_TABLES_INDEX.md. The coalescent simulations for the inversion analysis (section 2.2) were generated with msinv v0.3.5, a general ARG-based inversion-coalescent simulator (<https://github.com/stsmall/msinv>, MIT); this study’s complete *Illex illecebrosus*-specific inversion analysis — fitting code, simulation configuration, tabulated outputs, and the inversion figures — is contained within that repository under files/illex/ and results/illex/.

Acknowledgements

[TODO: funding, sequencing, sample collection, computing.]

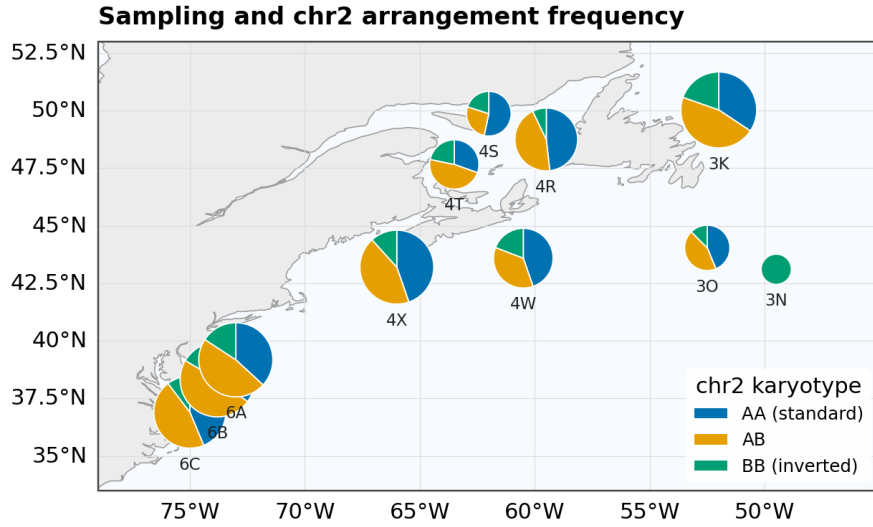


Figure 1: **Sampling and chromosome-2 arrangement frequency.** *Illex illecebrosus* sampled across eleven NAFO divisions ($\sim 37\text{--}50^\circ\text{N}$); pies show the AA/AB/BB karyotype composition at each division (sized by N). The inverted (BB) frequency is non-clinal ($0.29\text{--}0.46$ across latitude), consistent with geographic $F_{ST} \approx 0$ — the inversion is not locally adaptive.

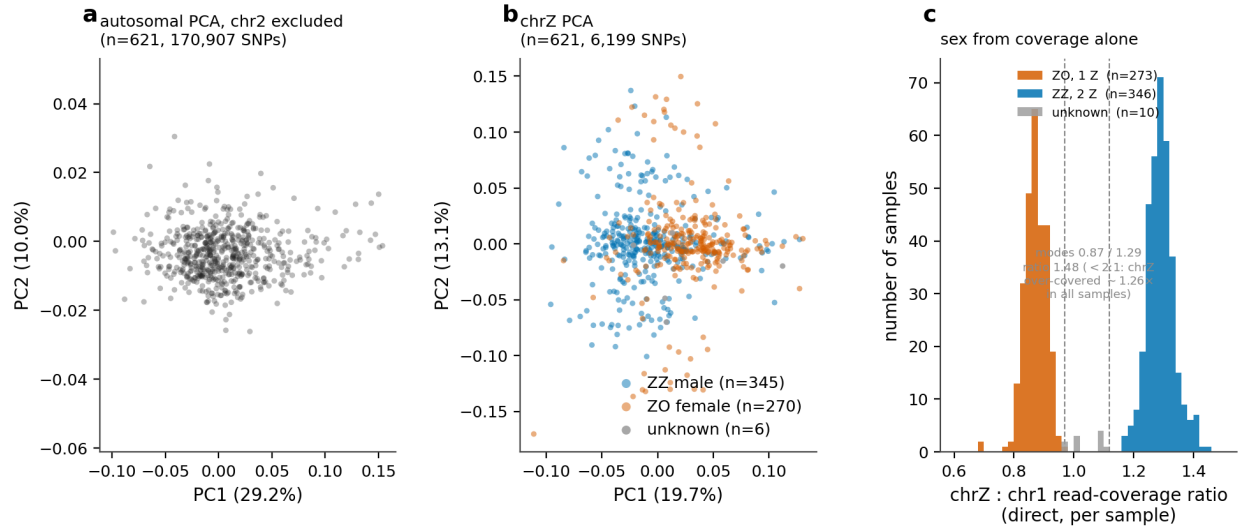


Figure 2: **No population structure outside chr2, and the sex assignment** (consistent with the range-wide panmixia of Baker et al. 2025). (a) Genome-wide autosomal PCA with chr2 excluded — a single undifferentiated cloud. (b) chrZ PCA (mild male/female separation). (c) chrZ:chr1 read-coverage ratio by individual: ZZ males at ~ 1.0 (two Z copies), ZO females near the single-copy expectation (there is no W); individuals in the ambiguous band are left unassigned. 342 males / 291 females.

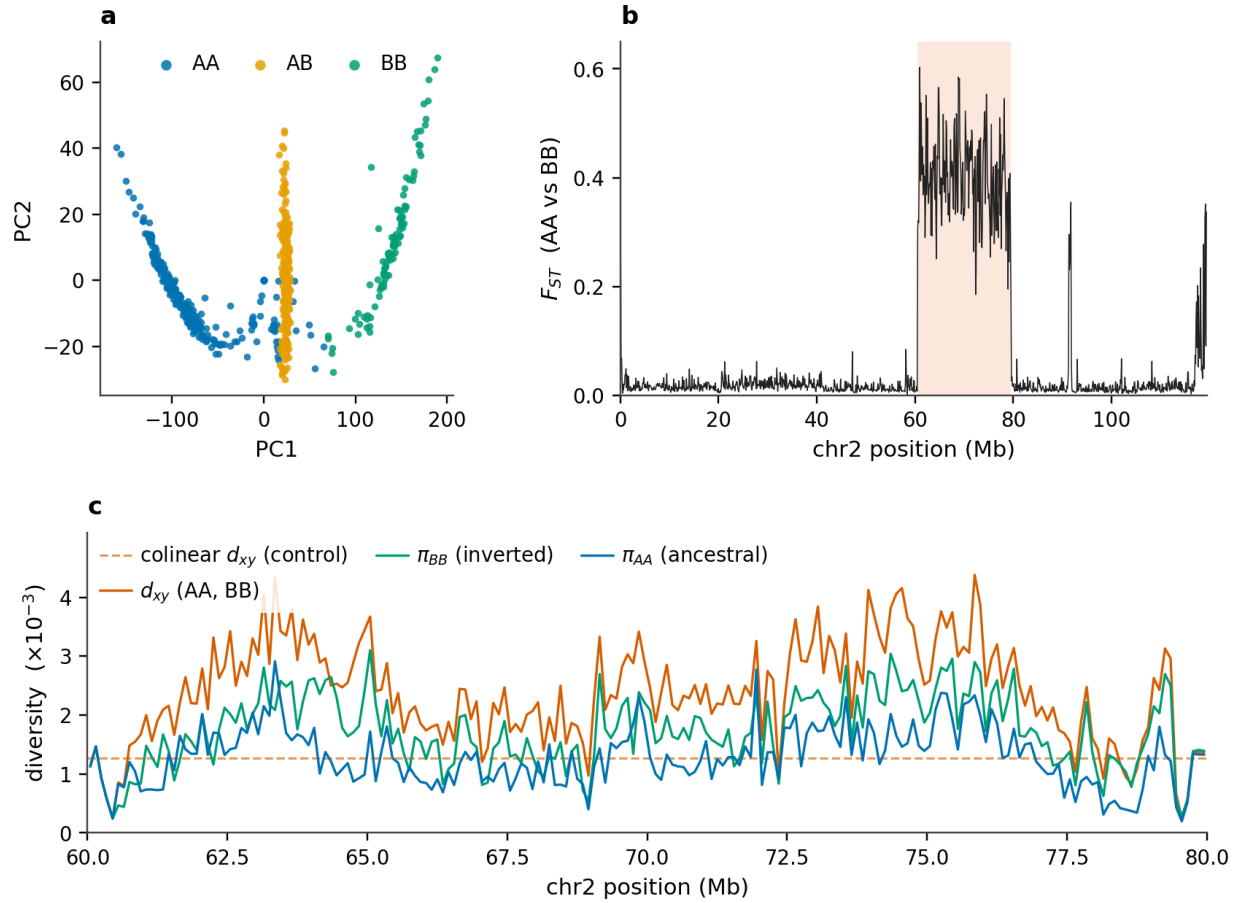


Figure 3: **The chromosome-2 inversion.** (a) chr2 PCA resolves three karyotype clusters (AA/AB/BB). (b) F_{ST} between arrangements along chr2 — the inversion (60.54–79.50 Mb, shaded) is the only differentiated region (median $F_{ST} = 0.37$ vs 0.01 collinear; whole-chromosome view in Supplementary Fig. S20). (c) Per-arrangement diversity across the inversion: the derived, inverted BB arrangement is *more* diverse than the ancestral AA ($\pi_I/\pi_S = 1.36$) despite its lower frequency, and d_{xy} exceeds both — the signature of an old, formerly common polymorphism.

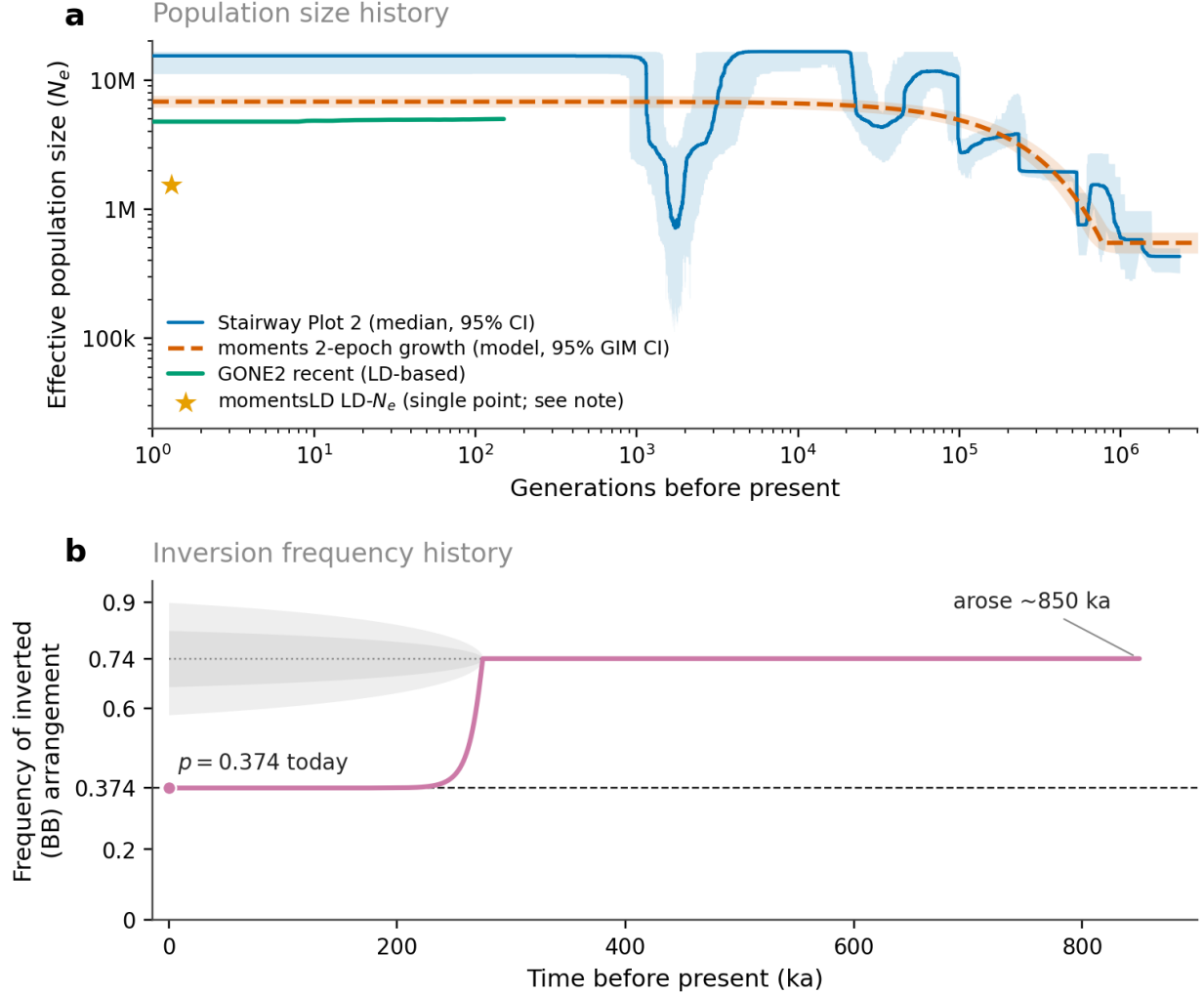


Figure 4: **Demographic history and the inversion's frequency history.** (a) Effective population size over time: Stairway Plot 2 (median $\pm 95\%$ CI), the moments 2-epoch growth model with its Godambe (GIM) 95% parameter-CI envelope (expansion $12.4 \times \pm 9\%$, $T \approx 770$ ky $\pm 16\%$), GONE2 for the recent past, and a single momentsLD LD- N_e point (rate-corrected; only the contemporary point is shown because its time-varying fit is non-identifiable). (b) Inversion trajectory fit with the barrier-aware coalescent simulator msinv: the derived (BB) arrangement arose ~850 ka — coincident with the onset of expansion in (a) — was held near $p \approx 0.74$, and declined to 0.374; the neutral-drift envelope is shown for comparison. $\Delta\chi^2$ profiles locating the end and duration of the decline are in Supplementary Fig. S14; the fall was recent and fast, excluding neutral drift-out.

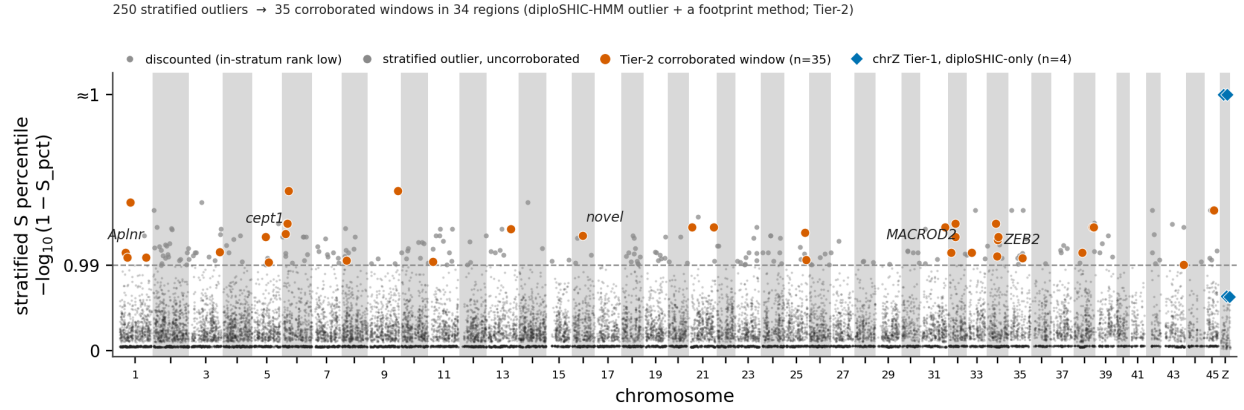


Figure 5: **The genome-wide selective-sweep scan.** Sweep signal across the genome (chromosomes concatenated). The classifier reads the genome as pervasively *linked-selected* ($\sim 91\%$ linked-hard/soft; neutral $\sim 1\%$; true hard/soft only a few percent), so raw diploS/HIC calls are filtered by cross-method concordance (a diploS/HIC-HMM outlier plus RAiSD or SweepFinder2) and a background-selection test; the corroborated, background-selection-robust candidates (35 windows across 34 regions; highlighted) stand out above the discounted background, with key genes labelled. Chromosome-Z Tier-1 calls (diamonds) are male-based and diploS/HIC-only — the footprint methods do not cover the Z — so they cannot enter the concordance filter. Chromosome 2 carries no concordant sweep. The full scan pipeline is shown in Supplementary Fig. S8.

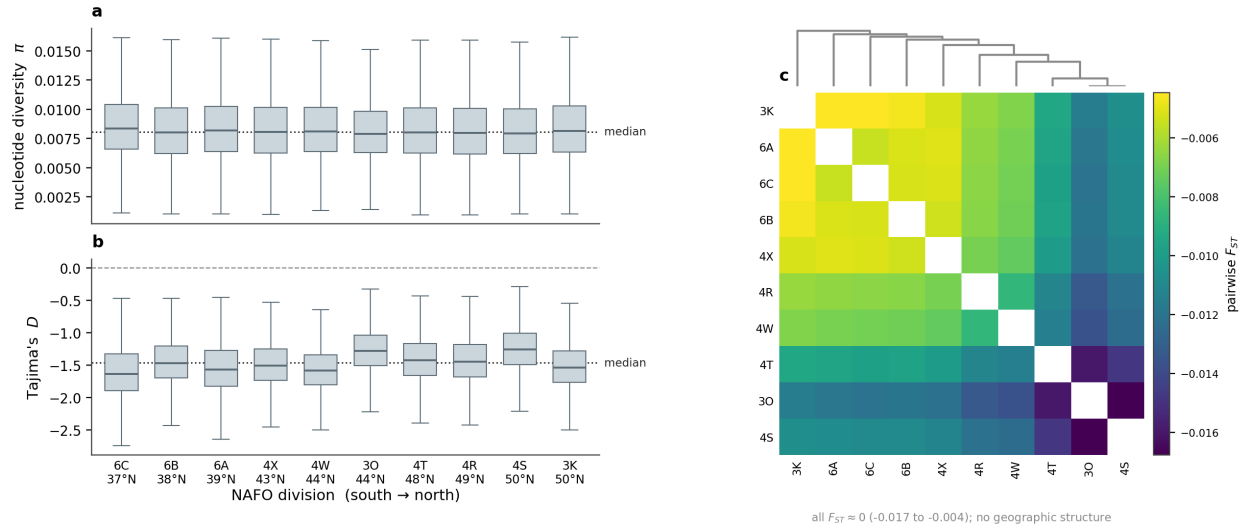


Figure 6: **Diversity is uniform and there is no geographic structure.** (a) Nucleotide diversity π and (b) Tajima's D per NAFO division, ordered south→north — both are flat across latitude (no cline), and D is negative in every population, the signature of strong expansion. (c) Pairwise F_{ST} among divisions as a clustered heatmap with dendrogram: every value is ≈ 0 (-0.017 to -0.004) and the clustering follows sample size, not geography — no population structure. (π/D estimated by ANGSD genotype likelihoods over all 45 autosomes plus the Z, large divisions subsampled to $n = 25$; F_{ST} genome-wide by pg_gpu. Per-chromosome distributions: Supplementary Fig. S9.)

Table 2: **The chromosome-2 inversion** (corrected polarisation: I = BB inverted/derived, S = AA ancestral/standard).

Quantity	Value
Location (differentiated body)	chr2:60.54–79.50 Mb (~ 18.96 Mb)
Karyotypes (AA/AB/BB)	254 / 284 / 95 (95.3% confident)
Inverted (BB) frequency p	0.374
Historical peak frequency	~ 0.74
Karyotype F_{ST} (AA,BB)	0.365 (geographic $F_{ST} \approx 0$)
π_I/π_S (I=BB)	1.356 ± 0.048
d_{xy}/π_I	1.385 ± 0.021
Coalescent age t_{inv}	~ 850 ky ($\propto 1/\mu$, $\mu = 3 \times 10^{-9}$)
Decline	$0.74 \rightarrow 0.374$ over ~ 50 – 100 ky, ending $\gtrsim 50$ – 100 ka
Selection on decline	$N_e s \sim 10^2$ – 10^3 ($s \approx 1.5 \times 10^{-4}$)
Gene content	89 gene models, 45 annotated; no GO enrichment ($p = 0.51$)
Fixed differences (AA vs BB)	3,513; 19 exonic, in 9 genes

Table 3: **Demographic model** (moments, folded SFS; validated by forward simulation).

Parameter	Value
Ancestral N_e (N_REF)	547,928
Present N_e (N0)	6,808,096
Growth duration (T_GROW)	769,519 generations
Fold expansion	$\sim 12\times$
Genome-wide π	0.0093
Genome-wide Tajima’s D	≈ -2 (nearly all windows)
Mutation rate μ	3×10^{-9} / bp / generation
Recombination (autosomal)	~ 0.21 – 0.24 cM/Mb (sex-averaged)

Table 4: **High-confidence selective-sweep candidates**, grouped by biological function and ordered by chromosome, including the Z-chromosome set. Autosomal rows are the 11 background-selection-robust candidates (of 34 concordant regions); Z rows (marked) are the fine-scale-confirmed sweeps from the males-only scan. Genes are cephalopod orthologs. Class: hard/soft sweep. Full machine-readable tables: Supplementary Table S6.

Location	Gene / feature	Class
<i>Neural, RNA-regulation & development</i>		
chr35:39.0 Mb	<i>ZEB2</i> (neural/EMT transcription factor)	hard
Z:20.8 Mb	<i>Pasilla</i> /Nova-1 (splicing regulator)	hard (Z)
Z:23.6 Mb	<i>Smaug</i> /samd4a (translational repressor)	hard (Z)
<i>Cardiovascular</i>		
chr1:20.3 Mb	<i>Aplnr</i> (apelin receptor)	hard
<i>Genome-maintenance & stress</i>		
chr32:9.3 Mb	<i>MACROD2</i> (DNA-damage response)	soft
<i>Membrane & lipid</i>		
chr5:38.0 Mb	<i>cept1</i> (phospholipid synthesis)	hard
chr1:85.5 Mb	ceramide phosphoethanolamine synthase [†]	hard
<i>Metabolism</i>		
chr25:65.0 Mb	<i>GALT</i> [†]	soft
<i>ECM & adhesion</i>		
chr33:9.0 Mb	<i>Tinagl1</i>	hard
<i>Signalling</i>		
chr32:24.1 Mb	GPCR cluster	hard
Z:30.7 Mb	<i>Esrrg</i> (estrogen-related receptor)	soft (Z)
<i>Cephalopod-specific novelty</i>		
chr16:35.3 Mb	uncharacterised (<i>Euprymna</i> family)	hard
<i>Lower confidence (TE-dense)</i>		
chr1:35.9 Mb	transposable element [‡]	hard
chr11:10.8 Mb	transposable element [‡]	hard

[†] target reassigned by fine-scale localisation (annotation-recovered gene). [‡] TE-dense region; deep- π /negative- D may be a TE-insertion sweep or a repeat artifact.

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